PROJECT PROPOSAL · ADP FY 2026-27

Bangladesh Before All

সবার আগে বাংলাদেশ

Kurigram Integrated Fertilizer Governance & Farmer Card Pilot (KIFGP) — A Cabinet Division Flagship Reform Initiative for Food Sovereignty and Farmer Dignity

Government of the People's Republic of Bangladesh

Cabinet Division · In coordination with the Ministry of Agriculture

Implementing Agency: Department of Agricultural Extension (DAE), Kurigram

Total Investment: Tk 3,949.6 lakh (≈ US$ 3.21 million) · Duration: 36 months

Project Code (proposed): MoA / DAE / KUR / 2026-27 / 01

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Executive Summary

This proposal places before the Cabinet Division a flagship reform initiative — the Kurigram Integrated Fertilizer Governance and Farmer Card Pilot (KIFGP) — that converts Bangladesh's Tk 17,001 crore annual fertilizer subsidy from a channel of chronic leakage into an instrument of precision, accountability and farmer dignity. The pilot is sited in Kurigram, a district that already produces a 537,543-tonne food surplus and carries the republic's hardest char geography, so that success here is proof of success anywhere.

The architecture is four layers — identity and land (the Farmer Card), point of sale (biometric, offline-first), tracking and transparency (a permissioned blockchain ledger with a public dashboard), and advisory (a land-based Soil Health Grid) — plus a Modified One-Must-One-Should extension package and a char resilience promise of 107 solar irrigation pumps, 9 speedboats and 9 fair-price collection centres. The operational ledger already exists as working code in the companion e-bill system repository; the reform's integrity layer is therefore not on the critical path of new development.

The total investment is Tk 3,949.6 lakh (approximately US$ 3.21 million) over 36 months, financed 60 per cent from the ADP, 25 per cent from PARTNER/World Bank co-finance, 10 per cent from PPP/private sources and 5 per cent in-kind. Funding is released in three annual tranches against two decision gates. In steady state the project recovers or saves approximately Tk 78.9 crore per year; the ten-year NPV at a 12 per cent discount rate is roughly Tk 260 crore of benefits against Tk 36 crore of costs, giving a benefit–cost ratio of 6–7:1 and an EIRR of approximately 35 per cent — decision-robust even at 50 per cent benefit realisation.

The proposal resolves to seven signatures requiring no new legislation and a single budget cycle: issue the biometric PoS circular within 90 days; complete 100 per cent Farmer Card enrolment within 12 months; link stock allocation to land data by Boro 2026-27; approve KIFGP at Tk 3,949.6 lakh in ADP FY 2026-27; adopt the land-based Soil Health Grid; launch Modified One-Must-One-Should; and sanction the char resilience and market package. The Cabinet Division is asked not merely to approve a district project but to endorse KIFGP as the national reference architecture for fertilizer governance, to be replicated district by district once its two decision gates are cleared in Kurigram.

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1. Introduction and the Cabinet Division Mandate

This proposal places before the Cabinet Division a flagship reform initiative that converts Bangladesh's fertilizer subsidy system from a channel of chronic leakage into an instrument of precision, accountability and farmer dignity. The Kurigram Integrated Fertilizer Governance and Farmer Card Pilot (KIFGP) is presented as a single, financeable project that the Cabinet Division can sponsor across ministries, anchoring the Ministry of Agriculture as the implementing authority and drawing in the Local Government Division, the ICT Division, the Finance Division, the Planning Commission and the Implementation Monitoring and Evaluation Division (IMED) as co-implementers and oversight partners. It is framed deliberately as a Cabinet-level instrument because the reform's success depends on coordination that no single line ministry can deliver alone.

The Cabinet Division is the apex coordinating body of the Government of the People's Republic of Bangladesh. It alone can convene the Secretary of Agriculture, the Finance Division, the Planning Commission, IMED, the ICT Division and the Local Government Division around one decision table, mandate a common digital backbone, and bind the outcome to a public accountability cycle. This proposal therefore asks the Cabinet Division not merely to approve a district project, but to endorse KIFGP as the national reference architecture for fertilizer governance — a pilot whose every component is designed to be replicated, district by district, across the country's 64 districts once its two decision gates are cleared in Kurigram.

The initiative answers the manifesto commitment “Korbo Kaj, Gorbo Desh” line by line in a single district: a real Farmer Card in every farmer's hand, a Tk 10,000 agricultural-loan waiver administered on a verified mobile-financial-services rail, biometric point-of-sale integrity at the last mile, soil-test-based entitlements rather than blanket cards, and a char resilience package that brings the state back into the fields that never stopped planting. It is, in the framing of the briefing material, “walking the path towards nationalism for sovereignty — one verified kilogram, one dignified farmer, one reclaimed taka at a time.”

2. Problem Analysis: The Fertilizer Distribution Crisis

Bangladesh spends roughly Tk 17,001 crore annually on fertilizer subsidy — close to eighty per cent of all public agricultural expenditure. Yet, according to World Bank distribution analysis, the bottom forty per cent of farmers receive only about fifteen per cent of the subsidy benefit, while the top quintile captures roughly half. The warehouse is full; the farmer's hand is not. Between the two stands a layer of leakage: unrecorded dealer sales, premiums over the printed maximum retail price, diversion to neighbouring markets, and allocation that is negotiated through dealer networks rather than tied to land and crop.

2.1 Three symptoms observed in the field

Field evidence collected by DAE Kurigram and the PARTNER programme over the 2023–26 cycle documents three converging failures. First, price premiums over the official price: in Rajshahi a 50 kg bag of TSP priced officially at Tk 1,350 sells at Tk 1,700–1,800 (a 33 per cent premium), while in Jashore the per-kilogram price of TSP doubles from roughly Tk 26 to Tk 52. In Kurigram Sadar, only one-third of the August 2025 urea allocation was lifted within the season — a classic artificial-scarcity signal. Second, agronomic over-application born of guesswork: 95.2 per cent of surveyed Kurigram mustard farmers apply urea above the recommended dose, 94.8 per cent over-apply TSP and 89.6 per cent over-apply Muriate of Potash; 56.8 per cent decide their dose by personal experience and only 9.2 per cent consult a specialist. Third, institutional failure: dealer lifting of 33–45 per cent of allocation within season, unrecorded sales, black-market premiums, and subsidy leakage and diversion.

2.2 Why Kurigram first

Kurigram is not a deficit district. It is the district that turned monga into surplus. It produces a food balance surplus of 537,543 tonnes, has scaled BRRI dhan 100+ thirty-six-fold in two years, has 12,350 hectares of Boro rice under Alternate Wetting and Drying saving thirty per cent of irrigation water, holds farmer-owned savings of Tk 12.40 lakh across 51 Farmer Service Centres in Nageshwari, and exported 50 tonnes of mango in 2025-26 with a 150-tonne target. It also carries the hardest geography: sixteen rivers cross the district, 382 chars sprawl across 54,720 hectares, and flood water can rise fifty centimetres in twenty-four hours. If the reform works here, it works anywhere — and Kurigram has already done the hard part. Asia has already paid for the lessons.

3. Project Objectives and Expected Outcomes

The project pursues six objectives, each tied to a measurable outcome indicator and a verifiable data source, so that the Cabinet Division and IMED can evaluate progress against evidence rather than narrative.

1. Eliminate subsidy leakage and price distortion through fingerprint-match authentication at every fertilizer sale (point-of-sale terminal plus upazila local server), with crop-declaration and land-parcel validation.
2. Complete the manifesto's Farmer Card in Kurigram: 100 per cent enrolment of 566,202 farm families (508,493 families plus 57,709 sharecroppers) on the Krishak Card, linked to National ID and to BDS/EDLMS land parcels.
3. Replace blanket soil-card targets with a land-based Soil Health Grid: 11,087 composite tests generating 436,902 parcel-linked digital Soil Health Cards.

4. Raise nutrient-use efficiency and farmer income through the Modified One-Must-One-Should package — certified seed plus soil-test-based entitlement, with five crop-specific reductions.
5. Secure char agriculture: solar irrigation, extension access by water, fair-price collection centres, and stress-tolerant variety seed systems.
6. Institutionalise real-time and performance audit, open tendering, union-level single-window service, and independent oversight, in line with the 31-Point State Reform Plan and the July National Charter.

Table 3.1 Expected outputs and Year-3 outcome indicators

Output	Quantity	Year-3 outcome indicator
Biometric PoS terminals (107 outlets + 9 spares)	116	95% of sales authenticated
Upazila local servers + district mirror/ledger	9 + 1	PoS uptime ≥ 98% incl. char outages
Farmer Card enrolment (biometric + parcel linkage)	566,202 families	100% coverage; e-voucher ready
Soil grid composite tests (1/20 ha plain; 1/10 ha char; +10% QC)	11,087	436,902 parcel-linked digital SHCs active
Lead farmers trained (1 per 10 families; ≥50% women)	56,620	Excess nitrogen −30%
Modified One-Must-One-Should block demonstrations	1,314	Adoption ≥ 60% of demo practice
Solar irrigation pumps (1 per char block)	107	Rabi irrigation crisis resolved in 107 blocks
Speedboats for char extension access	9	Char field-visit frequency ×3
Fair-price produce collection centres	9	Mango export 50 → 150 t; post-harvest loss −2%
Char agroforestry saplings	100,000	Char bank stabilization

Source: KIFGP Development Project Proforma, Section 7. Beneficiary counts from DAE Kurigram research-extension workshop census.

4. Solution Architecture: Four Layers, One Grid, One Package

The proposed journey replaces the existing Import/BADC → Dealer → Farmer chain (validated informally by familiarity and a paper card) with Import/BADC → local server plus biometric PoS → farmer's hands. Authentication is a fingerprint match between the PoS terminal and the upazila local server, reinforced by crop declaration and land-parcel

validation against the BDS and EDLMS registries. Subsidy releases only on a verified sale, mirroring India's DBT-PoS model. Every kilogram is traced on a permissioned blockchain ledger with a public dashboard. The architecture is organised as four layers, one grid, one package and one char promise.

4.1 Layer 1 — Identity and Land (the Farmer Card)

One hundred per cent enrolment of 566,202 farm families on the Krishak/Farmer Card, biometrically linked to National ID and to BDS/EDLMS land parcels via Union Parishad single-window desks. Each farmer record carries 120 data points. The GPMS baseline of 0.02 lakh rises to 5.66 lakh, and the same backbone supports the national target of 22 million national IDs by 2028 and the Korea MoU on land digitisation.

4.2 Layer 2 — Point of Sale (fingerprint match, offline-first)

116 biometric PoS terminals are deployed across 107 outlets (76 union/paurashava dealers under the 2026 Dealer Policy plus 31 char hubs) with nine spares. Nine upazila local-server nodes plus a district mirror ensure that char sales never stop for a network outage. Authentication combines fingerprint match, crop declaration and land validation, with a Bangla voice interface so that the non-smartphone majority is never excluded.

4.3 Layer 3 — Tracking and Transparency (the real-time audit)

A Central Digital Ledger using permissioned blockchain traces every kilogram from import through warehouse, dealer and farmer. A public dashboard exposes the data; automated leakage alerts flag anomalies; quarterly public reports are submitted to the Manifesto Implementation Committee. The audit regime has three layers: real-time ledger audit, an annual independent performance audit, and open e-GP tendering for every procurement package.

4.4 Layer 4 — Advisory and Verification (the Soil Health Grid)

Rather than blanket cards, the project builds a land-based grid: 5,676 plain samples at one per 20 hectares, 4,403 char samples at one per 10 hectares, and 1,008 quality-control tests, totalling 11,087 lab tests that generate 436,902 parcel-linked digital Soil Health Cards for every landholding family. The national scale-up formula is the same 20-ha/10-ha grid — roughly 400,000–450,000 tests nationally — not millions of physical cards. SRDI performs full-profile tests at Tk 1,500; the digital card costs a marginal Tk 40; satellite NDVI provides verification; and USSD/SMS delivers Bangla advisory to non-smartphone users.

4.5 Modified One-Must-One-Should and the char promise

The extension instrument is a Modified One-Must-One-Should package, adapted from Vietnam's “One Must, Five Reductions” (which cut fertilizer use by 31.3 per cent and

nitrogen and irrigation water by 20 per cent). The “Must” is certified seed plus a soil-test-based, land-linked entitlement; the “Should” is five crop-specific reductions — seed rate, SHC-dosed fertilizer, AWD or solar water, IPM pesticide, and post-harvest loss through collection centres — tuned per rice, mustard, maize, wheat, potato and char vegetables. It is delivered through 1,314 block demonstrations (two per block per year over three years, inputs costed at market price), 56,620 lead farmers (one per ten families, at least half women), and 187 SAAOs retrained as site-specific advisors. The char promise adds 107 solar irrigation pumps (one per char block, ending the Rabi water crisis), 9 speedboats for field officers, 9 fair-price collection centres, 100,000 char saplings, and BRRI dhan 111/114 community seed for flood-prone blocks.

5. Implementation Roadmap and Decision Gates

The project runs for 36 months, from 1 July 2026 to 30 June 2029, aligned with the DAE Kurigram GPMS three-year plan and the PARTNER programme cycle. Funding is released in three annual tranches against two decision gates, so that the Cabinet Division and IMED retain a clear go/adjust/stop lever at the end of each year rather than committing the full envelope upfront.

5.1 Decision gates

Gate 1, at the end of Year 1 (July 2027), requires verified 100 per cent Farmer Card enrolment and at least 95 per cent transaction capture at live outlets; meeting both releases the Year-2 tranche. Gate 2, at the end of Year 2 (June 2028), requires leakage-detection alerts to be operational, at least 50 per cent parcel-to-Soil-Health-Card linkage, and at least 60 per cent PoS sales share; meeting all three releases the Year-3 tranche. An independent baseline survey is conducted at Month 3 and an endline at Month 33, both by a competitively selected firm.

5.2 Asian precedent evidence

The design is de-risked by four neighbouring precedents. India's biometric PoS network covers 287,000 retailers and 84 crore farmers with no denial since 2018, cancelled 5,000-plus licences in seven months, and budgets ₹1.71 lakh crore for 2026-27 — and its e-Bill system ties payments to verified dealer sales, exactly the mechanism this project adopts. Vietnam's One Must, Five Reductions cut fertilizer use by 31.3 per cent. The Philippines delivered ₱5,000 cash to 600,000 rice farmers via verified mobile transfers. China reached over 99 per cent of villages with fibre or 4G, driving a 3.5 per cent efficiency gain for connected farmers, and adopted a 2015 zero-growth fertilizer action plan. Each precedent is mapped to a specific Kurigram component.

6. Resource Requirements and Budget

The total project cost is Tk 3,949.6 lakh (Tk 39.50 crore, approximately US$ 3.21 million at Tk 123 to the dollar) over 36 months, built bottom-up from Kurigram's actual geography. Every line item carries a unit, a count and a unit rate; there are no lump sums and no magic numbers. The capital share is 20.6 per cent (PoS terminals, servers, solar pumps, speedboats, collection centres); the balance is recurrent development expenditure. A 10 per cent contingency for physical and price variation (Tk 359.1 lakh) is included.

Table 6.1 Component cost build-up (Tk lakh)

#	Component	Cost (Tk lakh)
C1	Digital infrastructure — 116 PoS @ Tk 45,000; 9 upazila servers @ Tk 3.5 lakh; district mirror + ledger node	101.7
C2	Software stack (dealer app w/ Bangla voice, USSD/SMS farmer app, officer dashboard, policy engine) + 2-yr O&M	220.0
C3	Farmer Card enrolment — 566,202 families × Tk 100	566.2
C4	Soil Health Grid — 11,087 tests × Tk 1,500 + 436,902 digital cards × Tk 40	341.1
C5	Capacity building — 56,620 lead farmers × Tk 900; 187 SAAO × Tk 8,000; 214 dealers × Tk 5,000	535.3
C6	Modified One-Must-One-Should demos — 1,314 × Tk 25,000 at market-rate inputs	328.5
C7	Advisory comms — 9.06M SMS/USSD msgs/yr × Tk 0.35 × 3 yr	95.1
C8	Char resilience — 107 solar SIP @ Tk 3.5 lakh; 9 speedboats @ Tk 10 lakh; 100k saplings @ Tk 60	524.5
C9	State-managed collection centres — 9 × Tk 25 lakh (shed, weighbridge, cold room)	225.0
C10	e-voucher MFS gateway — 436,902 × Tk 15 × 2 Boro seasons	131.1
C11	Governance — oversight & public hearings; 3rd-party performance audit; baseline + endline	191.0
C12	PMU — 14 staff × Tk 55,000/mo × 36; office/vehicle	331.0
	Subtotal	3,590.5
	Contingency 10% (physical + price)	359.1
	TOTAL — 36 months (≈ US$ 3.21M @ Tk 123)	3,949.6

Source: KIFGP DPP Section 15 (Components) and Annex G (Unit-rate build-up).

6.1 Financing mix

The financing mix is 60 per cent ADP (Government development budget, Tk 2,369.8 lakh), 25 per cent PARTNER/World Bank co-finance (Tk 987.4 lakh), 10 per cent PPP/private including dealer co-pay and MFS corporate social responsibility (Tk 395.0 lakh), and 5 per cent in-kind contribution from BADC, BCIC, SRDI and DAE (Tk 197.5 lakh). Approval is not contingent on self-funding; the manifesto mandate and the fiscal-integrity case stand independently, and the co-finance is structured to bridge any delay in ADP release.

Table 6.2 ADP annual phasing (Tk lakh)

Component	Y1 2026-27	Y2 2027-28	Y3 2028-29	Total
C1 Digital infrastructure	101.7	—	—	101.7
C2 Software + O&M	140.0	40.0	40.0	220.0
C3 Farmer Card enrolment	450.0	116.2	—	566.2
C4 Soil Health Grid	178.6	120.0	42.5	341.1
C5 Capacity building	180.0	200.0	155.3	535.3
C6 One-Must-One-Should demos	110.0	110.0	108.5	328.5
C7 Advisory comms	30.0	33.0	32.1	95.1
C8 Char resilience	200.0	226.9	97.6	524.5
C9 Collection centres	50.0	100.0	75.0	225.0
C10 e-voucher MFS	—	65.5	65.6	131.1
C11 Governance & audit	65.0	63.0	63.0	191.0
C12 PMU & logistics	110.4	110.3	110.3	331.0
Subtotal	1,615.7	1,184.9	789.9	3,590.5
Contingency 10%	161.6	118.5	79.0	359.1
Annual total	1,777.3	1,303.4	868.9	3,949.6

Y1 carries 45% (Gate 1: 100% enrolment + 95% transaction capture). Y2 carries 33% (Gate 2: leakage alerts live + 50% parcel-SHC linkage). Y3 carries 22%.

7. Institutional Architecture and Governance

The project is governed by a five-tier architecture that deliberately avoids creating a parallel bureaucracy. Existing DAE offices host the field coordinators; existing Union Parishad desks host the single-window service; existing SRDI labs host the soil grid. The Cabinet Division's role is convening and accountability, not execution.

- Steering Committee — chaired by the Secretary, Ministry of Agriculture; meets quarterly; approves gate reviews. Members: Finance Division representative, Director General DAE, Manifesto Implementation Committee liaison, and a Cabinet Division observer.
- Project Management Unit (PMU) — at DAE Kurigram, with the Deputy Director as Project Director; 14 staff (PD, 2 deputies, 2 M&E, 3 IT, 2 accounts, 4 field coordinators). Upazila field coordinators are embedded in existing DAE offices.
- Union Parishad single-window desks — for Farmer Card enrolment, Soil Health Card download, grievance intake and e-voucher assistance. Local problems are resolved locally, in line with the manifesto.
- Oversight Committee — farmer leaders, civil society, UP chairs and technical experts; produces quarterly public transparency reports feeding the Manifesto Implementation Committee's public accountability assessments.
- Technical partners — SRDI (labs), BARI/BRRI (variety pipeline), BADC (dealer enforcement under the 2026 Policy), a2i and the ICT Division (cloud and UID-grade reader standards).

Funds flow from the ADP release to a PMU designated account over the iBAS/CGA rails; partner co-finance flows via the programme account; in-kind contributions are logged at framework rates. Every payment event is recorded on the ledger; monthly reconciliation with the BADC MIS; quarterly unaudited financial reports to the Finance Division; annual audit by the Comptroller and Auditor General plus an independent performance audit (component C11).

8. Risk Analysis and Mitigation

Eight risks have been identified with likelihood, impact and a named owner. None is treated as a reason to defer approval; each is paired with a concrete mitigation already costed into the components above.

Table 8.1 Risk register

#	Risk	L / I	Mitigation	Owner
1	Flood/char access disrupts rollout	H / M	Speedboats, offline-first PoS, raised-plinth centres	Project Director
2	Network failure at outlet	M / H	9 local servers, sync-on-reconnect, uptime trigger ≥90%	IT lead
3	Dealer resistance / rent-seeking	M / H	2026 Policy enforcement, licence review	BADC
4	Enrolment data quality	M / M	NID match + parcel validation + 10% field audit	PMU

#	Risk	L / I	Mitigation	Owner
5	Women's participation shortfall	M / M	Women-only batches, local venues, day-time scheduling	Gender focal
6	Fiscal delay in ADP release	M / M	Phased gates, partner co-finance bridge	PD / Finance
7	Political / policy discontinuity	L / H	Anchoring in manifesto four pillars + MIC reporting	Secretary MoA
8	Asset maintenance lapse post-exit	M / M	Maintenance clauses, user-group tariffs	DAE

L = Likelihood, I = Impact (H/M/L). Source: KIFGP DPP Section 23.

9. Economic and Financial Analysis

The project is justified as a public investment, and confirmed by arithmetic. In steady state (Year 4 onward) the project recovers or saves approximately Tk 78.9 crore per year: Tk 36.0 crore from subsidy-leakage recovery at 10 per cent of the district subsidy flow, Tk 32.4 crore from farmer input-cost savings at 15 per cent savings and 60 per cent adoption, Tk 9.1 crore from post-harvest loss reduction through nine collection centres, and Tk 1.4 crore from diesel displaced by 107 solar irrigation pumps.

Table 9.1 Steady-state annual benefits (Year 4+)

Benefit	Tk crore / yr
Leakage recovery @ 10% of subsidy flow	36.0
Farmer input-cost saving (15% @ 60% adoption)	32.4
Post-harvest loss cut via 9 collection centres	9.1
Diesel displaced by 107 solar SIPs	1.4
Total ≈ Tk 78.9 crore / yr	78.9

Source: KIFGP DPP Section 21. Subsidy flow through Kurigram ≈ Tk 360 crore/yr (120,000 t/yr).

9.1 Returns and sensitivity

Over a ten-year horizon at a 12 per cent discount rate, benefits approximate Tk 260 crore against costs of approximately Tk 36 crore. The benefit–cost ratio is approximately 6–7:1 and the economic internal rate of return approximately 35 per cent. Under a stress test at 50 per cent benefit realisation, the BCR remains approximately 3:1 — decision-robust rather than marginal. Payback on leakage recovery alone is under two steady-state years. The one-time Tk 39.50 crore outlay equals roughly 11 per cent of a single year's district subsidy flow; per farm family it is Tk 697 over three years (about Tk 232 per family per year, or US$

5.67 per family over three years), against documented per-bigha savings of Tk 1,052.35 per bigha per year — a savings-to-cost ratio of roughly 9.4:1 per bigha per year.

10. Monitoring, Evaluation and Audit Framework

Monitoring is evidence-led and IMED-compliant. Quarterly progress and financial reports follow the IMED template; an annual project evaluation is conducted; a completion report is submitted within twelve months of closure. An independent performance audit (component C11) is conducted annually on an outcome basis, not a compliance basis. A public transparency dashboard and quarterly public hearings at union level ensure that accountability runs downward to the farmer as well as upward to the Cabinet Division.

Table 10.1 Logical framework — results indicators

Indicator	Baseline	Year 2	Year 3	Data source
% fertilizer sales via biometric PoS	0%	60%	95%	Central Digital Ledger
Farmer Card enrolment	0.02 lakh	5.66 lakh	maintained	Krishak Card DB
Parcel-linked digital SHCs	0	218,451	436,902	SRDI grid
Subsidy share to bottom 40%	15%	30%	40%+	MFS / e-voucher data
Stock lifted ≤ 15 days	33–45%	85%	95%	BADC MIS
Subsidy leakage	High	<10%	<5%	Audit + blockchain trace
Excess nitrogen reduction	TBD	−15%	−30%	Soil + water tests
Women's share in training	28–39%	45%	50%+	Training registers
Char blocks with solar irrigation	0	72	107	DAE field reports
AWD area	12,350 ha	20,000 ha	25,000 ha	IPM / AWD monitoring
Farmer transaction time	a morning	<10 min	<10 min	PoS logs + survey
Grievance disposal	90%	95%	98%	Grievance portal

Source: KIFGP DPP Section 20 (Logical Framework & MEL Results Framework).

11. The Digital Governance Stack

Layer 3 of the architecture (tracking and transparency) is not a paper specification. It is already engineered. The companion software repository to this proposal — the e-bill system — carries a Prisma data model purpose-built for fertilizer subsidy governance, with the following entities already defined and database-ready: a User model with National ID hash, KYC status and verification tier (FARMER, DEALER, ADMIN); a Dealer model with licence number and geofencing (latitude, longitude, service radius in kilometres); a Fertilizer model with composition, subsidy rate (percentage) and a per-farmer-per-season entitlement cap; a Transaction model carrying farmer and dealer, a SHA-256 transaction hash for blockchain integrity, geofence-verified latitude and longitude, subsidy amount, farmer payment, a blockchain-recorded flag and a QR-code reference; a SoilHealthCard model with N/P/K/pH/organic-matter levels and recommended-fertilizer JSON; an InventoryItem model; and an AuditLog model capturing every action, entity, user, old/new values and IP address.

In other words, the operational ledger that the Cabinet Division is being asked to mandate already exists as working code: it needs deployment, not invention. Existing API routes already cover dealers, fertilizers, customers, bills and a dashboard. The reform's "Central Digital Ledger with permissioned blockchain" corresponds to the Transaction model's transactionHash and blockchainRecorded fields; the "public dashboard" corresponds to the existing dashboard route; the "automated leakage alerts" correspond to the AuditLog anomaly layer; and the QR-code reference field corresponds directly to the verified QR receipt each farmer receives at the point of sale. This is why the project can credibly target 95 per cent PoS-authenticated sales by Year 3: the integrity layer is not on the critical path of new development.

The combined public repository accompanying this proposal (cabinet-division-bangladesh-before-all) holds four source repositories in its reference-repos directory — the e-bill system (the operational subsidy ledger), the fertilizer reform portfolio site (the read-only briefing catalog with nine published documents and an inbox-driven site builder), the fertilizer-reform single-page reform package, and the fertilizer-nationalism briefing and Development Project Proforma from which this proposal is derived. Together they form a single professional-grade repository: a strategic communications portal (GitHub Pages), a formal proposal (this document), and a deployable digital governance stack.

12. Sustainability and Exit Strategy

Exit is designed from day one so that no project asset becomes a stranded liability. Five arrangements carry the project's outputs into steady-state operations: the PoS and server operation and maintenance is absorbed into DAE revenue budget and dealer licence fees; the Central Digital Ledger is transferred to the Ministry of Agriculture ICT wing; the 107 solar irrigation pumps are handed to water-user groups operating a volumetric tariff; the nine collection centres are operated by the Farmer Service Centres converted into cooperatives

or SMEs (the 51 FSCs in Nageshwari holding Tk 12.40 lakh in savings are the seed of this conversion); and the lead-farmer network continues within the GPMS framework. The e-voucher and MFS gateway, once proven, becomes the default rail for the manifesto's Tk 10,000 agricultural-loan waiver.

13. Conclusion and the Seven Signatures

The proposal resolves to seven signatures requiring no new legislation and a single budget cycle. Each action unlocks the next, and together they convert the manifesto's promise into a verified, auditable, farmer-facing reality in one district, on a backbone designed to scale nationally.

1. Issue the biometric PoS circular. The Secretary of Agriculture mandates biometric PoS piloting in all divisions within 90 days; Kurigram first, on the Krishak Card and NID backbone.
2. Complete 100 per cent Farmer Card enrolment in Kurigram. 566,202 families within 12 months through Union Parishad single-window desks.
3. Link stock allocation to BDS/EDLMS land data. BADC and DAE quotas are land-linked and crop-linked by Boro 2026-27, not negotiated through dealers.
4. Approve KIFGP at Tk 3,949.6 lakh in ADP FY 2026-27. 36-month phasing with two gates; open e-GP tendering; 60/25/10/5 financing mix.
5. Adopt the land-based Soil Health Grid. 11,087 composite tests and 436,902 parcel-linked digital cards; retire blanket card targets nationally.
6. Launch Modified One-Must-One-Should. 1,314 market-priced block demos; 56,620 lead farmers (at least 50 per cent women); FSC-to-cooperative conversion.
7. Sanction the char resilience and market package. 107 solar SIPs, 9 speedboats, 9 collection centres; independent oversight with quarterly public reports and a performance audit.

The ultimate measure of fertilizer security is not how much fertilizer Bangladesh holds in its warehouses. It is whether a farmer can obtain the right fertilizer, at the right price, at the right place, at the right time. This project is the Cabinet Division's instrument to make that true — first in Kurigram, then across the republic.

Annex A. District Profile — Kurigram

Kurigram district covers 2,245 square kilometres (224,500 hectares) with a population of 2,387,675 (male 1,179,695; female 1,207,980). It comprises 9 upazilas, 73 unions, 3 paurashavas, 219 blocks and 187 Sub-Assistant Agriculture Officers. Gross cropped area is 343,383 hectares and net cropped area 157,553 hectares, giving a cropping intensity of 217 per cent. The single, double, triple and quadruple cropped areas are 18,513 / 93,525 / 44,390 / 1,125 hectares respectively, with 153 hectares fallow. Land classification: high

47,302 ha, medium-high 72,875 ha, medium-low 30,304 ha, low 6,797 ha, very low 275 ha. The district spans Agro-Ecological Zones 2, 3, 7, 8, 9 and 22. Food balance 2024-25: production 821,653 tonnes minus demand 284,109 tonnes equals a surplus of 537,543 tonnes. Cultivable land per family is 0.43 hectares.

Annex B. Beneficiary Table

Category	Count	Basis
Farm families (excl. sharecroppers)	508,493	DAE research workshop census
Sharecroppers	57,709	DAE research workshop census
Total Farmer Card enrolment	566,202	Component C3
Landless families	71,591 (14.07%)	livestock/horticulture pathways
Landholding families receiving digital SHC	436,902	Component C4
Marginal (35.85%)	182,319	targeting weight
Small (40.58%)	206,382	targeting weight
Medium (9.00%)	45,708	targeting weight
Large (0.49%)	2,493	targeting weight
Lead farmers / dealer staff / SAAOs trained	56,620 / 214 / 187	Component C5
Direct jobs created	≈450	PoS techs 116, block agents 219, SIP caretakers 107, PMU 14

Marginal and small farmers together receive 76.4% of targeting weight.

Annex C. Procurement Plan

Package	Type	Method	Tk lakh	Year
Biometric PoS terminals (116)	Goods	NCT (e-GP), STQC-grade readers	52.2	Y1
Servers, networking, solar backup (10 nodes)	Goods	NCT	49.5	Y1
Software development + O&M	Services	QCBS (local vendors; source-code escrow to GoB)	220.0	Y1–Y3
Enrolment field services	Services	NCT services / NGO co-contracting with UPs	566.2	Y1–Y2
Soil sampling & lab testing	Services	Direct contracting to SRDI (framework rate)	166.3	Y1–Y3

Package	Type	Method	Tk lakh	Year
Solar irrigation pumps (107)	Goods	NCT with 5-yr maintenance clause	374.5	Y1–Y3
Speedboats (9)	Goods	NCT	90.0	Y1
Collection centres (9 civil works)	Works	NCT works	225.0	Y2–Y3
Demos, training, communications	Operating	DAE SOP / community participation	958.9	Y1–Y3
MFS gateway	Services	Framework among licensed MFS providers (lowest fee)	131.1	Y2–Y3
Performance audit & surveys	Services	QCBS independent firms	155.0	Y1–Y3

All packages through e-GP under PPR 2025; open tendering as default.

Annex D. Unit-Rate Build-Up (selected)

Item	Unit rate	Quantity	Total (Tk lakh)
Biometric PoS terminal	Tk 45,000	116	52.2
Upazila server node	Tk 350,000	9	31.5
District mirror + ledger node	Tk 1,800,000	1	18.0
Farmer Card enrolment	Tk 100	566,202	566.2
Soil composite test (full profile)	Tk 1,500	11,087	166.3
Digital SHC (parcel-linked)	Tk 40	436,902	174.8
Lead-farmer training	Tk 900	56,620	509.6
One-Must-One-Should demonstration	Tk 25,000	1,314	328.5
SMS/USSD message	Tk 0.35	27,200,000	95.1
Solar irrigation pump (2 HP + pipe)	Tk 350,000	107	374.5
Speedboat	Tk 1,000,000	9	90.0
Char sapling	Tk 60	100,000	60.0
Collection centre	Tk 2,500,000	9	225.0
MFS transaction fee	Tk 15	873,804	131.1
PMU staff-month	Tk 55,000	504	277.2

Annex E. Manifesto Alignment Matrix

The project delivers the manifesto commitment “Korbo Kaj, Gorbo Desh” line by line, instrument by instrument, in a single district.

Manifesto commitment	Instrument in KIFGP
Farmer Card for every farmer	100% enrolment of 566,202 families on Krishak Card, biometric + parcel-linked
Tk 10,000 agricultural-loan waiver (incl. interest)	Administered on the e-voucher / MFS platform (C10)
31-Point State Reform Plan	Biometric PoS, blockchain ledger, public dashboards, performance audit, open e-GP
July National Charter	Union-level single-window desks; grievance disposal target 98%
19 Points of Ziaur Rahman	Self-reliance: cooperative conversion of FSCs, water-user-group tariffs, lead-farmer network
Vision-2030 / $1 trillion by 2034	Sovereign digital rails (BDS, Krishak Card, NID) built first in Kurigram
Delta Plan 2100 / 8th FYP / SDGs	SDG 1, 2, 12, 13, 15 — solar irrigation, AWD, soil health, leakage reduction
DAE Kurigram GPMS 2026-29	GPMS targets linked to KIFGP contributions across six workstreams
Korbo Kaj, Gorbo Desh	Delivered line by line in Kurigram

Annex F. Certification and Submission

This proposal is certified from verified district data and the PARTNER programme 2023-26 evidence base. It is submitted, through the Ministry of Agriculture, to the Planning Commission's Project Evaluation Committee for evaluation and to the Finance Division for concurrence, for inclusion in the Annual Development Programme for the financial year 2026-27. Signature blocks: Project Director (Proposed) — Deputy Director, DAE Kurigram; Director (Planning), DAE; Additional Secretary (Planning), Ministry of Agriculture; Secretary, Ministry of Agriculture; Concurrence — Finance Division (Wing to be assigned).